

Rule-Based and LLM-Assisted Evaluation of Counterfactual Explanations for Income Prediction

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Abstract

This project studies the evaluation of counterfactual explanations for income prediction. We use DiCE to generate counterfactuals on the Adult Income dataset and evaluate them with a deterministic metrics-only reference system. We then test whether a calibrated single LLM can approximate this reference and whether a second LLM call can provide useful human-readable explanations. The calibrated single LLM reaches high issue-label agreement with the reference, while the explainability layer adds qualitative value without changing the fixed verdicts.

Introduction and Problem Formulation

This project connects two seminar themes: **Counterfactuals for Explainability and Fairness in Machine Learning** and **Orchestrate, Secure & Scale Enterprise GenAI Architecture**. It is related to the first seminar because we study counterfactual explanations for a binary income prediction model. Given a person whose income is predicted to be below \$50K, we use DiCE to generate counterfactual examples that would change the model prediction to above \$50K. The central question is not only whether these counterfactuals flip the model output, but whether they are good explanations. In the context of explainable AI, a useful counterfactual should be close to the original input, plausible, actionable, and sparse, meaning that it should change only a small number of features.

The project is also related to **Orchestrate, Secure & Scale Enterprise GenAI Architecture** because we implement an LLM-assisted evaluation pipeline. The seminar discussed generative AI workflows, orchestration of AI services, hallucination control, auditability, human validation, and trustworthy LLM systems. Our final pipeline follows this perspective by using a calibrated single-LLM evaluator and a separate explanation layer around a deterministic reference system.

The main research question addressed in this report is:

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Can a calibrated single LLM approximate a deterministic metrics-only reference for counterfactual quality evaluation, and can an explanation layer add useful human-facing interpretation without changing the verdict?

This question is relevant because counterfactual generation methods such as DiCE typically select counterfactuals using deterministic criteria. These include **validity**, which checks whether the generated counterfactual flips the model prediction; **proximity**, which favors small changes from the original input; **sparsity**, which favors changing few features; and **feasibility**, which can restrict which features are allowed to change. These metrics are necessary, but they do not fully capture whether a proposed change is realistic, fair, causally coherent, or convincing from a human perspective. We therefore test whether LLM-based evaluators provide additional useful assessment of the counterfactuals generated by DiCE. Since no external ground-truth labels exist for the quality of these counterfactual explanations, we use a deterministic metrics-only evaluator as a rule-based reference standard. This reference encodes our project-specific human judgment through explicit heuristics and thresholds over validity, sparsity, proximity, confidence, and plausibility checks. We then evaluate whether a single LLM can substitute for this reference at the issue-label level and whether an optional explanation layer makes the verdicts easier to inspect.

We evaluate this question on the OpenML Adult Income dataset, where the task is to predict whether a person's income is above or below \$50K.

Related Work

Taxonomy of the Research Area

Machine learning is increasingly used in high-impact domains such as healthcare, finance, and banking. This makes explainability important, because many useful models are difficult to interpret directly. The research area can be divided into three relevant families of methods:

- **Intrinsically interpretable models:** methods such as linear models or decision trees are easier to inspect because their decision process is relatively transparent.

However, they may be less flexible than more complex models.

- **Post-hoc counterfactual explanations:** these methods explain a model decision by asking what minimal changes to the input would have led to a different prediction. They are useful because they can explain individual decisions without requiring access to the internal structure of the model.
- **Counterfactual quality evaluation:** generated counterfactuals are usually assessed with metrics such as validity, proximity, sparsity, diversity, and feasibility. These metrics are useful but incomplete, because they do not always capture whether a recommendation is realistic, fair, or actionable in the real world.

This taxonomy positions our project in the third family. We do not only generate counterfactual explanations; we study how their quality should be evaluated. In particular, we use deterministic quality metrics and heuristics as a stable reference, then test whether an LLM can reproduce and explain that reference.

Seminal and Recent Works

Counterfactual explanations were introduced as a way to explain automated decisions without opening the black box of the underlying model (Wachter, Mittelstadt, and Russell 2017). Instead of describing all internal model mechanisms, a counterfactual explanation answers a local question: what would need to change in this input for the model to produce a different outcome? For a classification task, this means showing how an unfavorable prediction could be changed into a favorable one.

Later work extended this idea by emphasizing diversity and actionability, rather than validity alone. DiCE generates multiple diverse counterfactual alternatives for a given instance, giving users several possible paths toward the desired outcome instead of a single potentially unrealistic modification (Mothilal, Sharma, and Tan 2020). This is the counterfactual generation framework used in our project.

However, DiCE-style evaluation still relies heavily on numerical criteria. Validity checks whether the model prediction flips; proximity measures how close the counterfactual is to the original instance; sparsity measures how many features change; diversity measures variation among generated alternatives; and feasibility constrains which features may be modified. These criteria are necessary, but they can miss important semantic failures. For example, a counterfactual may be close in feature space while still proposing an impossible change, such as decreasing a person’s age from 30 to 25. Similarly, numerical metrics may not fully capture fairness, realism, or causal coherence.

Our project addresses this gap with a deterministic reference evaluator and an LLM-assisted explanation layer. Because there is no external ground truth for these quality labels, our metrics-only evaluator acts as the reference standard: it encodes human judgment through explicit rules over

metrics and issue-specific plausibility checks. The LLM is then assessed by how well it reproduces this reference and whether it can explain the result in clearer language. To make the audit operational, we define a small issue taxonomy covering implausible time-dependent changes, extreme working hours, unactionable capital shifts, too many simultaneous changes, and fragile counterfactuals; the next section describes how these labels are used in the evaluation pipeline.

Method

The original predictive task is binary classification on the Adult Income dataset, where the model predicts whether an individual’s income is at most or above \$50K. We train a Logistic Regression model inside an `sklearn` pipeline. This model is used for counterfactual generation because DiCE requires reliable class-probability outputs that can guide its optimization procedure.

Before generating counterfactuals with DiCE, we define a feature policy that freezes attributes that should not be modified: `fnlwgt`, marital status, relationship, race, sex, and native country. Counterfactuals are therefore not allowed to assign different values to these features, since doing so would imply unfair or unrealistic demands on the individual.

After training the predictive model and applying this feature policy, we generate counterfactual explanations with DiCE. For each individual with an unfavorable prediction, DiCE produces multiple candidate counterfactuals that aim to move the prediction to the favorable class. We keep the DiCE generation parameters at their default values and use the feature policy only to restrict which attributes may change.

For each generated counterfactual, we compute a first set of standard numerical quality metrics:

- **Validity:** whether the counterfactual reaches the favorable class.
- **Proximity:** how close the counterfactual remains to the original instance.
- **Sparsity:** how many features are changed.
- **Diversity:** how different the generated counterfactuals are from each other.
- **Counterfactual confidence:** the model probability assigned to the favorable class for the counterfactual.

These metrics are useful for ranking and filtering counterfactuals, and they are close to the criteria commonly used in DiCE-style generation. However, they remain purely numerical. They do not directly represent whether a counterfactual is plausible, actionable, or convincing from a human perspective. We therefore add a five-label issue taxonomy that provides a second evaluation layer. The taxonomy defines a fixed vocabulary of possible counterfactual failures:

- `implausible_time_dependent_change`: age decreases, education decreases, education increases with-

out sufficient age increase, or age/education becomes non-integer.

- `extreme_working_hours`: weekly working hours become dangerously high, implausibly low, or change by an unrealistic amount.
- `unactionable_capital_shift`: capital gain or capital loss changes in a financially unrealistic way.
- `too_many_changes`: too many actionable features are changed at the same time.
- `fragile_counterfactual`: the counterfactual only barely reaches the favorable class, with confidence close to the decision threshold.

Each label corresponds to a specific failure mode in the generated counterfactual. The previous label `inconsistent_work_profile` was removed from the active taxonomy because the deterministic rules did not detect it reliably and LLM outputs tended to overuse it. This reduces noisy labels and keeps all evaluation strategies aligned with the same vocabulary.

We operationalize the taxonomy through deterministic heuristic rules. These rules encode human-designed plausibility checks over the original instance, the counterfactual, and the numerical metrics. For example, they flag decreases in age, unrealistic changes in education or working hours, large unexplained capital shifts, excessive simultaneous modifications, and low-margin favorable predictions. The heuristic module returns the set of issue labels detected for each counterfactual. We also compute metric warnings, including `count_diversity_low`, which flags counterfactual sets with low diversity. These warnings inform severity but are not scored taxonomy labels.

Finally, we define a metrics-only deterministic evaluator. This evaluator does not use an LLM. It receives as input the generated counterfactuals, their standard numerical metrics, and the heuristic issue flags derived from the issue taxonomy. It outputs two judgments for each counterfactual:

- **Issue severity**: low, medium, or high, depending on the number and type of detected issues.
- **Overall assessment**: fair, ambiguous, or unfair, summarizing whether the counterfactual is acceptable as an explanation.

This deterministic evaluator acts as the rule-based reference against which the LLM-based evaluation strategies are compared.

Single-LLM substitution evaluator. The main LLM-based system is a single evaluator prompted to approximate the metrics-only reference. It receives the original profile, generated counterfactuals, feature changes, numerical metrics, heuristic issue evidence, and taxonomy instructions. The prompt is calibrated to:

- approximate the metrics-only reference rather than invent a new evaluation policy;
- start from the heuristic issue union `flagged_issues.union`;

- keep labels supported by deterministic heuristic evidence;
- avoid unsupported labels outside the active taxonomy;
- preserve `fragile_counterfactual` when it is present in case-level or counterfactual-level heuristic flags.

The output is a structured verdict with an overall assessment (fair, ambiguous, or unfair), detected issue labels, severity, confidence, and recommended action.

Explanation layer. In the final version, explainability is added as a second LLM call after the verdict is fixed. This layer receives the fixed single-LLM verdict and compact case evidence, then produces an `expert_explanation` for a non-expert reader. It is explicitly forbidden from changing issue labels, severity, assessment, confidence, or the recommended action. This separates the substitution task from the human-facing interpretation task.

Earlier multi-agent system. A multi-agent AutoGen debate with Prosecutor, Defense attorney, Expert witness, and Judge was implemented during the project. In the final Phase 2 framing, however, the central evaluation is the single-LLM substitution and explanation pipeline, because the goal is to measure approximation of the deterministic reference rather than to prove that debate is more correct.

Experimental Setup

Dataset

We evaluate the pipeline on the Adult Income dataset. The task is binary income prediction: each individual is classified as earning at most \$50K or more than \$50K per year.

Property	Description
Dataset	Adult Income
Instances	48,842
Input features	14 demographic, educational, occupational, financial, and working-time attributes
Prediction task	Binary income classification
Unfavorable class	Predicted income $\leq$ \$50K
Favorable class	Predicted income $>$ \$50K

Table 1: Dataset used in the experiments.

The main features include age, education, occupation, capital gain, capital loss, hours worked per week, and native country. For the evaluation stage:

- we select 10 individuals predicted in the unfavorable class;
- DiCE generates multiple counterfactuals for each selected individual;
- DiCE generation uses its default parameters, with the feature policy described in the Method section.

Baselines

We compare the LLM systems against the metrics-only evaluator. This evaluator is not an external ground truth, but it is the deterministic reference system used for substitution scoring.

- **Metrics-only reference:** deterministic evaluator based on numerical metrics and heuristic issue flags.
- **Uncalibrated single LLM:** one LLM produces a structured verdict from the same case evidence.
- **Calibrated single LLM:** the single-LLM prompt is constrained to approximate the metrics-only reference and preserve heuristic labels.
- **Calibrated single LLM with explanation:** the calibrated verdict is followed by a second LLM call that explains the fixed verdict.

Metrics

Evaluation focuses on agreement with the metrics-only reference:

- **Detection rate:** percentage of taxonomy issue labels from the metrics-only reference that are also identified by the evaluated system.
- **False-positive rate:** percentage of cases where the evaluated system adds issue labels absent from the reference.
- **Exact match rate:** percentage of cases for which the complete predicted issue-label set exactly matches the metrics-only reference set.

Implementation Details

The implementation is written in Python. Counterfactuals are generated with DiCE, and the LLM-based evaluators are implemented with AutoGen. The final Phase 2 experiments use Groq with `llama-3.1-8b-instant`.

Setting	Calibrated LLM	LLM + explanation
Framework	AutoGen	AutoGen
Provider	Groq	Groq
Model	llama-3.1-8b-instant	
Temperature	0.2	
Max tokens	400	700

Table 2: LLM configuration used in the final Phase 2 experiments.

The temperature is set to 0.2 to keep outputs conservative while avoiding fully deterministic generation. The final explainability run uses a larger completion budget because it adds a second-stage human-readable explanation. Earlier multi-agent experiments used a smaller 250-token budget to stay under Groq’s token limits, but the final Phase 2 evaluation focuses on the single-LLM substitution and explanation pipeline.

Results and Analysis

Table 3 summarizes the main experimental results.

System	Detect.	FP	Exact
Uncalibrated single LLM	84.0%	0.0%	60.0%
Calibrated single LLM	96.0%	0.0%	90.0%
LLM + explanation layer	96.0%	10.0%	80.0%

Table 3: Substitution scores against the metrics-only reference.

Prompt calibration substantially improves issue-label substitution. Detection increases from 84.0% to 96.0%, and exact match increases from 60.0% to 90.0%. This shows that the LLM performs best when it is instructed to follow the deterministic heuristic evidence rather than act as a free-form evaluator.

The final explainability run keeps the same 96.0% detection rate but has lower exact match, 80.0%, and a 10.0% false-positive rate. This is interpreted as normal LLM variability in a fresh run, not as the explanation layer changing the labels: the explanation is generated only after the verdict is fixed.

Discussion

The results support the Phase 2 framing of the project. The deterministic metrics-only evaluator is the stable backbone: it defines the reference issue labels and keeps the evaluation auditable. The calibrated single LLM can approximate this reference well at the issue-label level, but only when the prompt explicitly tells it to preserve deterministic heuristic evidence. This suggests that the LLM is more reliable as a structured substituter than as an independent judge.

The explanation layer provides the clearest added value. It does not improve the quantitative substitution scores, and it is not allowed to change the verdict. Instead, it translates the fixed structured assessment into language that is easier for a reader to understand. This is useful because the metrics-only reference is precise and reproducible, but its raw issue labels and thresholds are not always self-explanatory.

The main limitation is that the reference itself is rule-based. It reflects our designed heuristics, not an external human-labeled ground truth. In addition, LLM outputs remain variable across runs, as shown by the difference between the best calibrated run and the final explainability run. Future work could test a larger model, add human annotations, or calibrate the metric-warning thresholds such as `count_diversity_low`.

Conclusion

This report studied counterfactual explanation evaluation for income prediction. We generated counterfactuals with DiCE, evaluated them with a deterministic metrics-only reference, and tested whether a single LLM could approximate

and explain that reference. The final conclusion is that deterministic rules remain the evaluation backbone, while the calibrated LLM is useful as a low-cost substitution and explanation layer. Its strongest contribution is not superior correctness, but clearer human-facing interpretation of an auditable rule-based verdict.

References

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